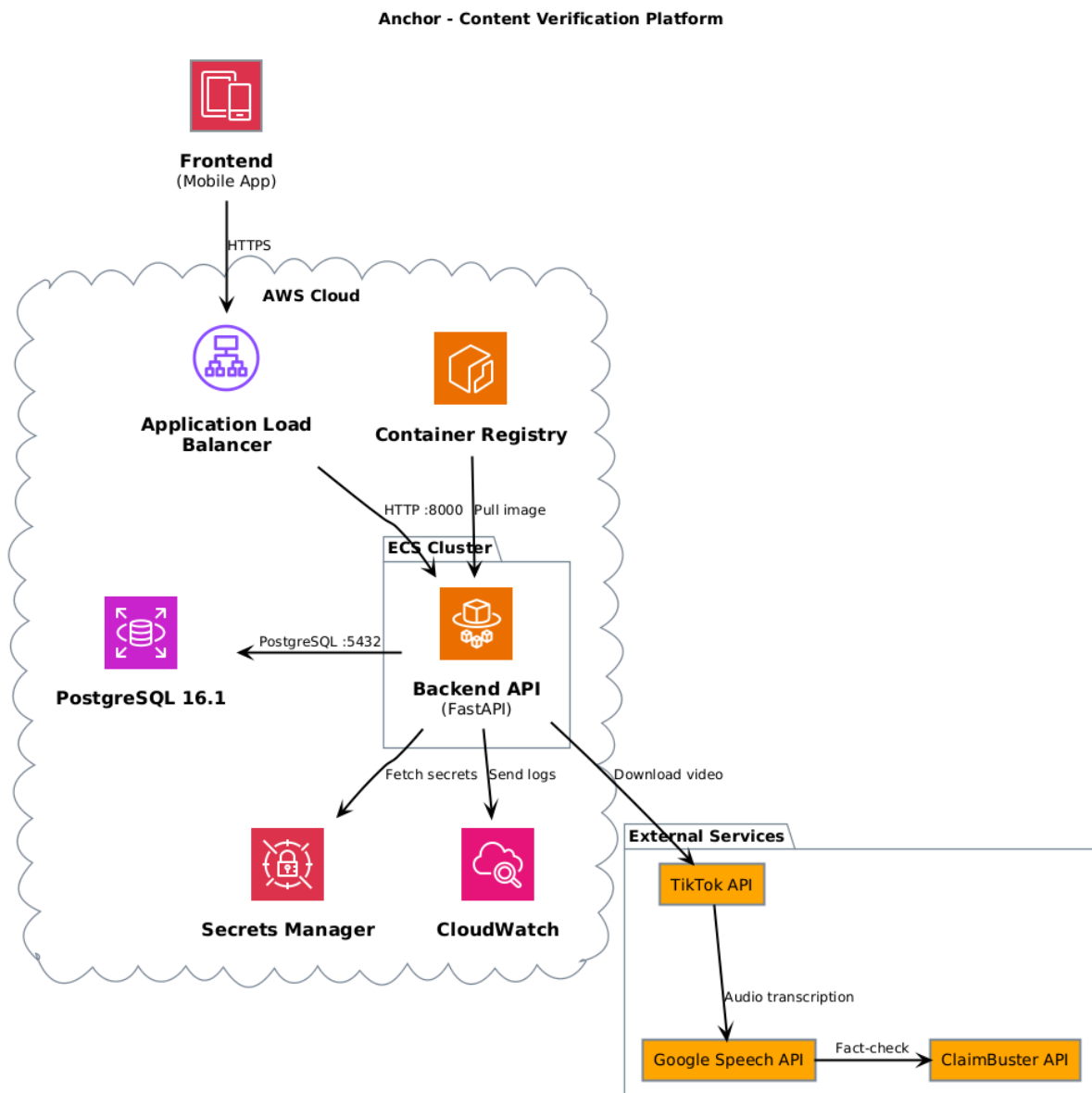


CSC490 Assignment 3 - Deleting Prod

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Part 1: Diagramming your Infrastructure



Current State:

The Anchor project analyzes problematic multimodal content on social media, focusing on TikTok. It includes:

- Authentication: User login/signup with JWT tokens (FastAPI + PostgreSQL).
- TikTok Analysis: Downloads TikTok videos using URL submission, extracts frames/audio, performs OCR, transcription, summarization, and fact-checking using external APIs.

Future State:

- Automated TikTok Content Fetching: Planned feature to periodically fetch and analyze TikTok content using the TikTok API. This will enable proactive detection of problematic content.
- Publish App to App/Play Store: Deploy app to app store to be downloaded for users.

Development vs. Production:

- Development:
 - PostgreSQL: Runs locally using a native installation.
 - Secrets: Managed via .env files
 - Postman for testing.
 - Backend: Runs locally using uvicorn with --reload for hot-reloading.
 - Frontend: Runs locally using Expo (React Native) with live reloading.
- Production:
 - AWS ECS Fargate: Stateless containers for the backend API.
 - AWS RDS PostgreSQL: Managed database with multi-AZ deployment for high availability.
 - Secrets Manager: Stores sensitive data like API keys and database credentials.
 - ALB (Application Load Balancer): Routes traffic to Backend ECS tasks.
 - Frontend (To do): Publish to App/Play Store and allow downloads on Mobile devices.
 - CloudWatch: Centralized logging and monitoring.
 - Terraform: Manages AWS infrastructure as code.
 - GitHub Actions: CI/CD pipeline for automated deployment.
 - Production URL: Backend runs behind an ALB with a public DNS.

Part 2: Infrastructure as Code Implementation

Terraform modules are created and kept at /infra on the github linked [[Terraform modules](#)]. Github workflow script has also been created to deploy the infrastructure to AWS.

Note: My AWS Innovation Box request has not yet been accepted, and my personal AWS account has run out of free credits, so I am unable to deploy this architecture currently.

Part 3: Deleting your Prod (exempted)